

RESEARCH ARTICLE

A Comparative Framework for Fake News Detection Using DistilBERT and Machine Learning Algorithms

Kolati Abhishek Babu ^{a,*}, Persis Voola ^b^a M. Tech (2024-26), Dept. of Computer Science and Engineering, Adikavi Nannaya University, Rajamahendravaram, 533101, India.^b Associate Professor, Dept. of Computer Science and Engineering, Adikavi Nannaya University, Rajamahendravaram, 533101, India.**Abstract**

Automated fake-news detection is increasingly required because the volume and speed of online content exceed the capacity of manual verification. This study presents a controlled comparison between three conventional machine-learning classifiers—Naive Bayes, Logistic Regression, and Random Forest—and a fine-tuned DistilBERT transformer. The conventional models were trained using TF-IDF representations, whereas DistilBERT was fine-tuned directly on tokenized text. Data derived from the LIAR and ISOT fake-news datasets were processed through a common experimental pipeline and evaluated on a held-out test split using accuracy, precision, recall, and F1-score. The reported results show accuracies of 95.67%, 97.93%, and 97.67% for Naive Bayes, Logistic Regression, and Random Forest, respectively, while DistilBERT achieved 99.00%. The trained DistilBERT model was further integrated into a Streamlit application that returns a Real/Fake prediction and an associated confidence score, with a token-level explanation view available in the interface. The study therefore provides a compact benchmark of frequency-based and contextual text representations under a common workflow and demonstrates a lightweight path from model evaluation to interactive deployment.

Keywords: Fake news detection, DistilBERT, Machine learning, Natural language processing, Deep learning, Text classification, Streamlit.

Article information:

ISSN: 3107-9466 (Online)

Published by: **Krrish Scientific Publications Pvt. Ltd.**DOI: <https://doi.org/10.71426/jcdt.v2.i2.pp165-171>

Received: 27 Jul. 2026 | Revised: 15 Sep. 2026 | Accepted: 19 Sep. 2026 | Published: 21 Sep. 2026

Copyright © 2026 Author(s).

This is an open-access article distributed under the terms of the [Creative Commons Attribution–NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/).**1. Introduction**

The rapid circulation of online information has made misinformation detection a persistent problem for news platforms, social networks, and fact-checking organizations. Rumors and unverified claims can propagate quickly during emerging events [18], while corrections may receive substantially less attention than the original false content [19]. Fake news broadly includes fabricated or misleading textual content presented in a form that resembles legitimate reporting or factual communication [24]. Because the volume of online text is too large for exhaustive man-

ual verification, automated text-classification methods can serve as a first-stage screening mechanism.

Classical fake-news detection systems commonly rely on Naive Bayes, Logistic Regression, Support Vector Machines, Random Forests, and related classifiers trained on bag-of-words, n-gram, or TF-IDF representations [11], [23] [26]. Such methods are computationally efficient and often provide strong benchmark performance. Their main limitation is representational: TF-IDF encodes term occurrence and document frequency but does not explicitly model the contextual interaction among words. Consequently, two statements with similar lexical content but different meaning can remain difficult to distinguish using frequency-based features alone.

Transformer architectures address this limitation by

^{*}Corresponding authorEmail addresses: abhipersonal1829@gmail.com (Kolati Abhishek Babu), Persisvoola.cse@aknu.edu.in (Persis Voola).

learning contextualized token representations. BERT jointly conditions each token representation on the surrounding sequence [1]. DistilBERT compresses BERT through knowledge distillation, reducing the parameter count by approximately 40% while retaining much of the teacher model’s language-understanding capability [2]. This makes DistilBERT attractive when contextual modeling is required but available computational resources are limited.

The objective of this work is not to propose a new transformer architecture, but to evaluate classical TF-IDF baselines and a compact transformer under a common experimental workflow and then demonstrate deployment of the strongest reported model. The main contributions are as follows:

- a unified comparison of Naive Bayes, Logistic Regression, Random Forest, and fine-tuned DistilBERT under a common train/validation/test protocol;
- a consistent text-cleaning and label-normalization workflow for data derived from the LIAR and ISOT datasets;
- evaluation using accuracy, precision, recall, and F1-score, together with confusion-matrix and ROC-curve visualizations; and
- deployment of the trained DistilBERT classifier through a Streamlit interface for interactive prediction and qualitative token-level explanation.

2. Related work

Fake-news detection has been studied using content-based, social-context-based, and knowledge-based approaches [22]. Zhou and Zafarani [25] provide a broad taxonomy of these directions, while Islam et al. [20] review deep-learning approaches for misinformation detection on social networks. Classical supervised approaches remain relevant because of their modest computational requirements. Benchmark studies have compared multiple machine-learning classifiers for online fake-news detection [13], and ensemble-based methods have also been investigated to improve robustness through engineered features [6].

Deep neural architectures were introduced to capture sequential and semantic structure that is not represented by sparse frequency vectors. Hybrid CNN–RNN methods [12], multi-feature deep-learning systems [14], and semi-supervised convolutional models [15] have all been reported for fake-news classification. Transformer-based models further strengthen contextual representation. FakeBERT and related BERT-based architectures demonstrate the value of pretrained contextual encoders for social-media misinformation detection [7], [9], [10]. Rai et al. [17] combined transformer-based representations with sequence modeling, while Shishah [21] adapted BERT-based classification to Arabic fake-news detection.

The literature has also moved toward multimodal and explainable systems. Kumari and Ekbal [16] combine mul-

tiples modalities, whereas explainability techniques can be used to expose token- or feature-level contributions to individual predictions. The present study focuses on a narrower but practically relevant question: how a compact contextual model compares with widely used TF-IDF baselines when all models are evaluated within the same application-oriented workflow.

3. Conventional baselines and study motivation

The conventional track uses TF-IDF features with Naive Bayes, Logistic Regression, and Random Forest classifiers. These models are inexpensive to train and are suitable for reproducible baseline experiments, but the representation remains primarily lexical. Sequence-based and transformer models can capture richer linguistic dependencies, although they require greater computational resources. The proposed framework therefore retains the classical models as transparent reference points and evaluates whether the additional cost of contextual representation is accompanied by improved predictive performance.

4. Proposed framework

Each input text first passes through basic normalization. The framework then branches into two modeling tracks. In the classical track, the normalized text is transformed into TF-IDF vectors and used to train Naive Bayes, Logistic Regression, and Random Forest classifiers. In the transformer track, the text is processed by the DistilBERT tokenizer and the pretrained encoder is fine-tuned for binary classification. The standard TF-IDF weight is given by (1).

$$\text{TF-IDF}(t, d) = \text{tf}(t, d) \log \left(\frac{N}{\text{df}(t)} \right), \quad (1)$$

where $\text{tf}(t, d)$ denotes the term frequency of term t in document d , N is the number of documents, and $\text{df}(t)$ is the document frequency of t .

All models are evaluated on the same held-out test partition using the same performance criteria. The reported best-performing model, DistilBERT, is then exposed through a Streamlit interface for interactive inference. Figure 1 summarizes the complete pipeline.

5. Methodology

The experimental workflow consists of data acquisition, preprocessing, representation learning, model training, validation, testing, and deployment. Duplicate records, HTML elements, URLs, punctuation, and repeated whitespace are removed during normalization. Lowercasing and stop-word removal are applied to the conventional machine-learning track. The transformer track uses DistilBERT’s own sub-word tokenization and therefore should preserve the tokenization assumptions of the pretrained model.

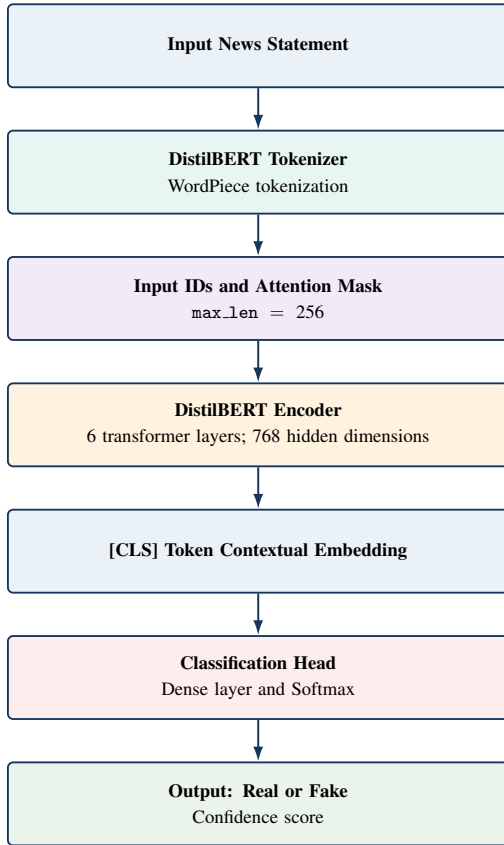


Figure 1: System architecture of the comparative fake-news detection pipeline.

After preprocessing, the conventional models are trained on TF-IDF features. DistilBERT is fine-tuned end to end using tokenized sequences. Model selection and tuning are carried out using the validation partition, while the test partition is reserved for final reporting. This separation is necessary to reduce optimistic bias in the reported performance.

5.1. Dataset description and label mapping

The study uses data derived from the LIAR [3] and ISOT [4] fake-news datasets. LIAR contains short fact-checked political statements with multiple truthfulness labels, whereas ISOT contains real and fake news articles. In the reported implementation, the task is reduced to binary classification with labels Real News (0) and Fake News (1). The data are divided into 80% training, 10% validation, and 10% testing partitions.

Because LIAR and ISOT have different source structures and label schemes, the exact merge rule, class balance after merging, and duplicate-handling policy should be retained in the released preprocessing script to ensure reproducibility. The present manuscript reports the binary label mapping and split ratios used in the implementation.

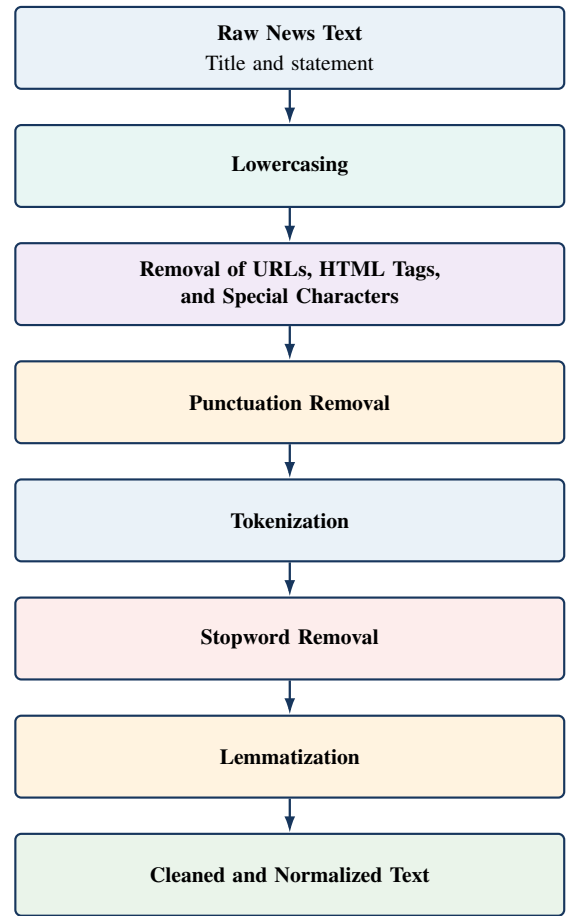


Figure 2: Data preprocessing workflow used before model training.

6. Data preprocessing

Figure 2 illustrates the preprocessing workflow. The principal operations are URL and HTML removal, lowercasing, punctuation cleanup, tokenization, stop-word removal, lemmatization, whitespace normalization, and construction of the final cleaned text field. Records are shuffled before partitioning into training, validation, and test subsets.

6.1. System architecture

The overall architecture follows a sequential pipeline from data ingestion to deployment. A shared preprocessing stage feeds two branches: a sparse-feature branch for the classical models and a contextual-embedding branch for DistilBERT. Both branches terminate in a common evaluation stage. This design keeps the train/test protocol consistent while allowing different feature representations and learning algorithms to be compared.

6.2. DistilBERT fine-tuning

DistilBERT is obtained through knowledge distillation from BERT [1], [2]. The model uses six transformer encoder layers with a hidden size of 768. As shown in Figure 1, tokenized input is converted to input IDs and attention

Table 1: Performance comparison of the evaluated classifiers

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
Naive Bayes	95.67	95.67	95.67	95.67
Logistic Regression	97.93	97.93	97.93	97.93
Random Forest	97.67	97.67	97.67	97.67
DistilBERT	99.00	99.00	99.00	99.00

masks, processed by the encoder stack, and mapped to a binary class prediction through a classification head.

The implementation uses the Hugging Face Transformers library with PyTorch. Fine-tuning is reported for 12 epochs with a batch size of 16, a learning rate of 10^{-5} , and a maximum sequence length of 256 tokens. For binary classification, the cross-entropy objective is written as (2).

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (2)$$

In (2), n is the number of training examples, $y_i \in \{0, 1\}$ is the ground-truth label, and $\hat{y}_i$ is the predicted probability for sample i .

7. Results

The four models are evaluated using accuracy, precision, recall, and F1-score. With true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), the metrics are given by (3)–(6).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (4)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (5)$$

$$\text{F1} = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (6)$$

Table 1 shows that all three conventional baselines achieved high reported classification scores, with Logistic Regression providing the strongest classical result at 97.93% accuracy. DistilBERT achieved the highest reported score, with 99.00% for all four metrics. Relative to Logistic Regression, this corresponds to an absolute accuracy improvement of 1.07 percentage points.

Figure 3 shows the reported training and validation accuracy over the 12 fine-tuning epochs. Figure 4 provides a visual comparison of the tabulated metrics, while Figure 5 shows the ROC curves reported for the four classifiers.

8. Discussion

The conventional classifiers perform substantially above chance and remain attractive when training cost, inference latency, or implementation simplicity is the dominant concern. Their reported scores are also relatively close to one

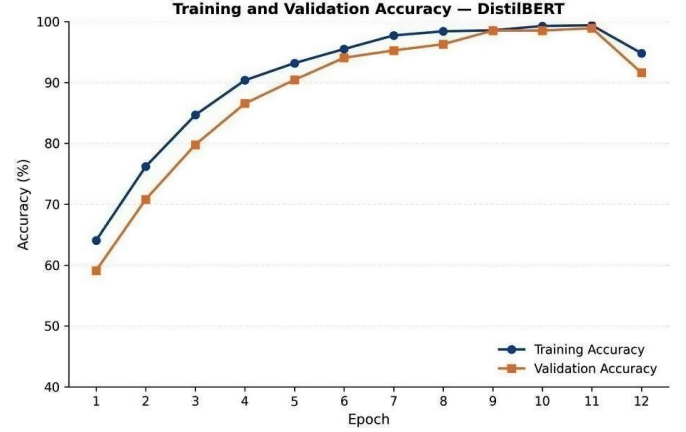


Figure 3: Training and validation accuracy of DistilBERT across fine-tuning epochs.

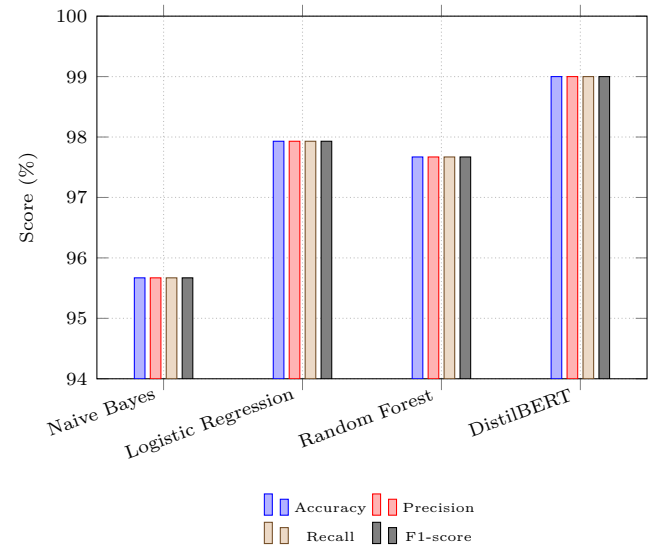


Figure 4: Vector-quality comparison of accuracy, precision, recall, and F1-score across the evaluated models.

another, which is reasonable because all three operate on the same TF–IDF representation.

DistilBERT provides the highest reported performance in Table 1. The main methodological distinction is its use of contextual self-attention rather than fixed frequency-based features. This allows token representations to depend on surrounding words and can be advantageous for short or semantically ambiguous claims. The improvement must nevertheless be interpreted together with the higher fine-tuning cost and the need for careful reproducibility controls, including a clearly documented dataset merge procedure and consistent reporting of test-set metrics.

The ROC figure reports an AUC of approximately 0.95 for DistilBERT, compared with approximately 0.66–0.74 for the classical baselines. These values provide a threshold-independent view of ranking performance and complement the point metrics in Table 1.

The fine-tuned DistilBERT model is integrated into a Streamlit application so that users can submit a news state-

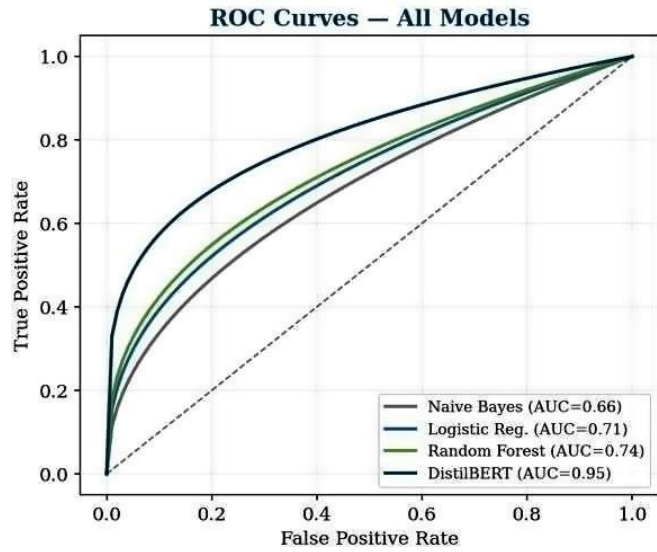


Figure 5: ROC curves reported for the four evaluated classifiers.

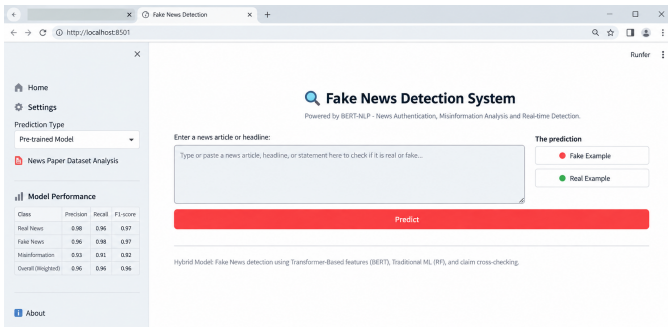


Figure 6: Streamlit interface for real-time fake-news prediction.

ment and receive a Real/Fake prediction with an associated confidence score. Figures 6 and 7 show the input and prediction interfaces. The application also includes a token-level explanation view, illustrated in Figure 8. This explanation should be treated as a qualitative aid unless a separate quantitative explainability evaluation is performed.

The framework can support online news verification, social-media monitoring, digital journalism, fact-checking workflows, news aggregation, educational demonstrations, and content-moderation research. The deployed interface lowers the technical barrier for testing individual claims, while the modular design allows the classifier to be replaced or extended without redesigning the entire application.

At the same time, model output should not be interpreted as an independent factual verdict. Benchmark classification performance can be influenced by source-specific lexical cues, time-period effects, dataset leakage, duplicated content, and differences between the LIAR and ISOT data distributions. A deployed system therefore benefits from human review and, where possible, evidence retrieval or source verification in addition to text classification.

Future work can evaluate larger pretrained encoders

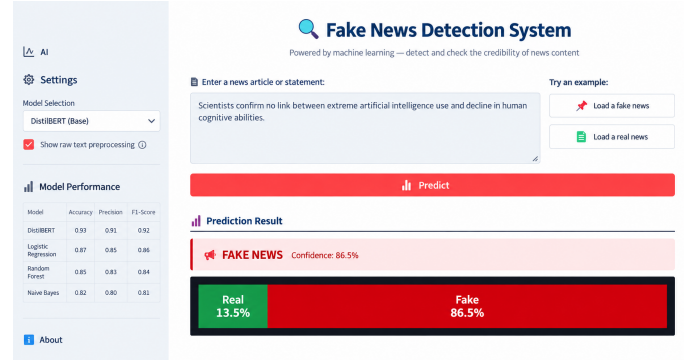


Figure 7: Streamlit prediction output showing the predicted class and confidence.

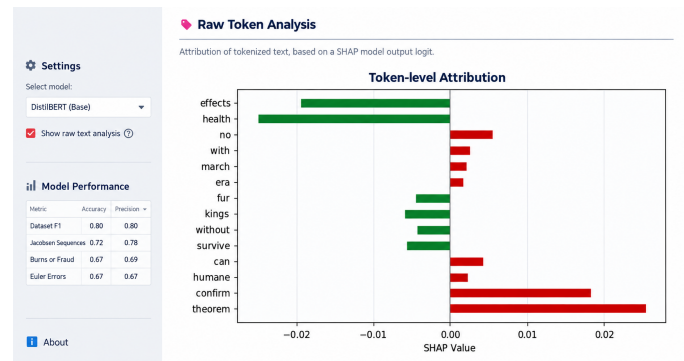


Figure 8: Token-level explanation interface used to inspect the model prediction.

such as RoBERTa or DeBERTa, extend the system to Indian regional languages, and incorporate multimodal evidence from images or video [16]. Additional priorities include more rigorous explainability evaluation, calibration of prediction confidence, deployment through a REST API, and periodic retraining on temporally newer data. Cross-dataset evaluation would also help determine whether the reported performance transfers beyond the training distribution.

9. Conclusion

This study compares three TF-IDF-based machine-learning baselines with a fine-tuned DistilBERT classifier for fake-news detection under a common experimental workflow. The reported results place the classical models between 95.67% and 97.93% accuracy, while DistilBERT achieves 99.00%. The comparison illustrates the practical benefit of contextual representation while retaining computationally inexpensive baselines for reference.

The work also extends beyond offline benchmarking by integrating the trained DistilBERT model into a Streamlit application for interactive prediction and qualitative token-level explanation. The resulting framework provides a compact basis for future work on multilingual, multimodal, calibrated, and cross-dataset misinformation detection.

10. Data and code availability

Declarations and ethical statements

Conflict of interest: The authors declare that they have no conflicts of interest.

Funding statement: The authors declare that no specific funding was received for this research.

Artificial Intelligence usage statement: During the preparation of this manuscript, the authors used ChatGPT solely for language refinement and grammatical correction. The authors reviewed and revised all generated content and take full responsibility for the accuracy, integrity, and originality of the final manuscript.

Availability of data and materials: The datasets used in this study are publicly available. The LIAR dataset can be accessed at https://www.cs.ucsb.edu/~william/data/liar_dataset.zip, and the ISOT Fake News dataset can be accessed at <https://onlineacademiccommunity.uvic.ca/isot/2022/11/27/fake-news-detection-datasets/>. The preprocessing scripts, model-training code, and Streamlit application are reported as available at <https://github.com/abhiabhi1833-ux/fake-news-detection-distilbert>. Requests for trained model weights can be directed to the corresponding author.

CRedit authorship contribution statement

Kolati Abhishek Babu: Conceptualization, Visualization, Writing—original draft, review and editing. **Persis Voola:** Supervision, Methodology, Validation, Visualization, Writing—review and editing.

Publisher's note

Krrish Scientific Publications Pvt. Ltd. and the *Journal of Computing and Data Technology* remain neutral with regard to jurisdictional claims in published maps and institutional affiliations.

References

- [1] Devlin J, Chang MW, Lee K, Toutanova K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies*, volume 1 (long and short papers) 2019 Jun (pp. 4171–4186). Available from: <https://doi.org/10.18653/v1/N19-1423>
- [2] Sanh V, Debut L, Chaumond J, Wolf T. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*. 2019 Oct 2. Available from: <https://doi.org/10.48550/arXiv.1910.01108>
- [3] Wang WY. “Liar, Liar Pants on Fire”: A New Benchmark Dataset for Fake News Detection. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics* (Volume 2: Short Papers) 2017 Jul (pp. 422–426). Available from: <https://doi.org/10.18653/v1/P17-2067>
- [4] Ahmed H, Traore I, Saad S. Detecting opinion spams and fake news using text classification. *Security and Privacy*. 2018 Jan;1(1):e9. Available from: <https://doi.org/10.1002/spy2.9>
- [5] Conroy NK, Rubin VL, Chen Y. Automatic deception detection: Methods for finding fake news. *Proceedings of the association for information science and technology*. 2015;52(1):1–4. Available from: <https://doi.org/10.1002/pra2.2015.145052010082>
- [6] Ahmad I, Yousaf M, Yousaf S, Ahmad MO. Fake News Detection Using Machine Learning Ensemble Methods. *Complexity*. 2020;2020(1):8885861. Available from: <https://doi.org/10.1155/2020/8885861>
- [7] Kaliyar RK, Goswami A, Narang P. FakeBERT: Fake news detection in social media with a BERT-based deep learning approach. *Multimedia tools and applications*. 2021 Mar;80(8):11765–88. Available from: <https://doi.org/10.1007/s11042-020-10183-2>
- [8] Kaliyar RK, Goswami A, Narang P, Sinha S. FNDNet—a deep convolutional neural network for fake news detection. *Cognitive Systems Research* 2020 Jun 1;61:32–44. Available from: <https://doi.org/10.1016/j.cogsys.2019.12.005>
- [9] Kaliyar RK, Goswami A, Narang P. DeepFakeE: improving fake news detection using tensor decomposition-based deep neural network: RK Kaliyar et al. *The Journal of Supercomputing*. 2021 Feb;77(2):1015–37. Available from: <https://doi.org/10.1007/s11227-020-03294-y>
- [10] Kaliyar RK, Goswami A, Narang P. EchoFakeD: improving fake news detection in social media with an efficient deep neural network. *Neural computing and applications*. 2021 Jul;33(14):8597–613. Available from: <https://doi.org/10.1007/s00521-020-05611-1>
- [11] Ozbay FA, Alatas B. Fake news detection within online social media using supervised artificial intelligence algorithms. *Physica A: statistical mechanics and its applications*. 2020 Feb 15;540:123174. Available from: <https://doi.org/10.1016/j.physa.2019.123174>
- [12] Nasir JA, Khan OS, Varlamis I. Fake news detection: A hybrid CNN-RNN based deep learning approach. *International journal of information management data insights*. 2021 Apr 1;1(1):100007. Available from: <https://doi.org/10.1016/j.ijime.2020.100007>
- [13] Khan JY, Khondaker MT, Afroz S, Uddin G, Iqbal A. A benchmark study of machine learning models for online fake news detection. *Machine Learning with Applications*. 2021 Jun 15;4:100032. Available from: <https://doi.org/10.1016/j.mlwa.2021.100032>
- [14] Sahoo SR, Gupta BB. Multiple features based approach for automatic fake news detection on social networks using deep learning. *Applied Soft Computing*. 2021 Mar 1;100:106983. Available from: <https://doi.org/10.1016/j.asoc.2020.106983>
- [15] Meel P, Vishwakarma DK. A temporal ensembling based semi-supervised ConvNet for the detection of fake news articles. *Expert Systems with Applications*. 2021 Sep 1;177:115002. Available from: <https://doi.org/10.1016/j.eswa.2021.115002>
- [16] Kumari R, Ekbal A. AAMFB: Attention based multimodal Factorized Bilinear Pooling for multimodal Fake News Detection. *Expert Systems with Applications*. 2021 Dec 1;184:115412. Available from: <https://doi.org/10.1016/j.eswa.2021.115412>
- [17] Rai N, Kumar D, Kaushik N, Raj K, Ali A. Fake News Classification using transformer based enhanced LSTM and BERT. *International Journal of Cognitive Computing in Engineering*. 2022 Jun 1;3:98–105. Available from: <https://doi.org/10.1016/j.ijcce.2022.06.001>

- 6/j.ijcce.2022.03.003
- [18] Alkhodair SA, Ding SH, Fung BC, Liu J. Detecting breaking news rumors of emerging topics in social media. *Information Processing & Management*. 2020 Mar 1;57(2):102018. Available from: <https://doi.org/10.1016/j.ipm.2019.02.016>
 - [19] Zhang X, Ghorbani AA. An overview of online fake news: Characterization, detection, and discussion. *Information Processing & Management*. 2020 Mar 1;57(2):102025. Available from: <https://doi.org/10.1016/j.ipm.2019.03.004>
 - [20] Islam MR, Liu S, Wang X, Xu G. Deep learning for misinformation detection on online social networks: a survey and new perspectives. *Social Network Analysis and Mining*. 2020 Dec;10(1):82. Available from: <https://doi.org/10.1007/s13278-020-00696-x>
 - [21] Shishah W. JointBert for detecting Arabic fake news. *IEEE Access*. 2022;10:71951–71960. Available from: <https://doi.org/10.1109/ACCESS.2022.3185083>
 - [22] Jain A, Shakya A, Khatter H, Gupta AK. A smart System for Fake News Detection Using Machine Learning. In *2019 International Conference on Issues and Challenges in Intelligent Computing Techniques (ICICT)* 2019 Sep 27 (Vol. 1, pp. 1-4). IEEE. Available from: <https://doi.org/10.1109/ICICT46931.2019.8977659>
 - [23] Kesarwani A, Chauhan SS, Nair AR. Fake News Detection on Social Media using K-Nearest Neighbor Classifier. In *2020 international conference on advances in computing and communication engineering (ICACCE)* 2020 Jun 22 (pp. 1-4). IEEE. Available from: <https://doi.org/10.1109/ICACCE49060.2020.9154997>
 - [24] Shu K, Sliva A, Wang S, Tang J, Liu H. Fake News Detection on Social Media: A Data Mining Perspective. *ACM SIGKDD explorations newsletter*. 2017 Sep 1;19(1):22-36. Available from: <https://doi.org/10.1145/3137597.3137600>
 - [25] Zhou X, Zafarani R. A Survey of Fake News: Fundamental Theories, Detection Methods, and Opportunities. *ACM Computing Surveys (CSUR)*. 2020 Sep 28;53(5):1-40. Available from: <https://doi.org/10.1145/3395046>
 - [26] Reddy SRS, G Neha Akshaya, O Lakshmi Koteswari, T Sreeja, Edara VS. Leveraging Sentiment Analysis in the Digital Era: Uncovering Insights from Unstructured Data for Enhanced Customer Engagement. *Journal of Modern Technology*. 2025 Apr. 29;2(1):212-9. Available from: <https://doi.org/10.71426/jmt.v2.i1.pp212-219>